7.5 *Neisseria Meningitides* (Meningococcus)

The genus *Neisseria* is included in the family *Neisseriaceae* (Figure 7.6). It contains two important pathogens *Neisseria meningitidis* and *Neisseria gonorrhoeae*, both the species are strict human pathogens. *N. meningitides* causes meningococcal meningitis (formerly known as cerebrospinal fever).

The word **Meningitis** is derived from Greek word **'meninx'** means membrane and **'itis'** means inflammation. It is an inflammation of meanings of brain or spinal cord. Bacterial meningitis is a much more severe disease than viral meningitis.

7.5.1 Morphology

They are Gram negative diplococci (0.6µm–0.8µm in size) arranged typically in pairs, with adjacent sides flattened.

They are non – motile, capsulated (Fresh isolates).

Cocci are generally intracellular when isolated from lesions (Figure 7.7).

7.5.2 Cultural Characteristic

They are strict aerobes, but growth is facilitated by 5–10% CO_2 and high humidity. The optimum temperature is 35°C–36°C and optimum pH is 7.4–7.6. They are fastidious pathogens, growth occurs on media enriched with blood or serum. They grow on the following media and show the characteristic colony morphology (Table 7.4).

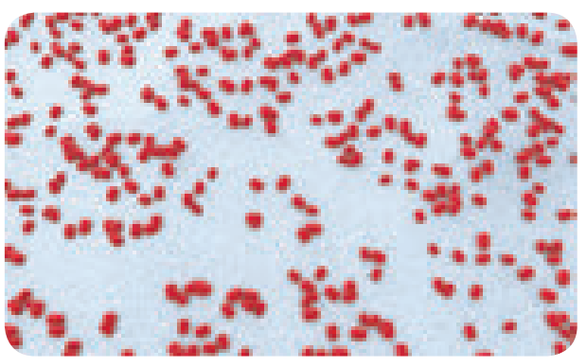


Figure 7.6: Gram staining of *Neisseria Meningitides*

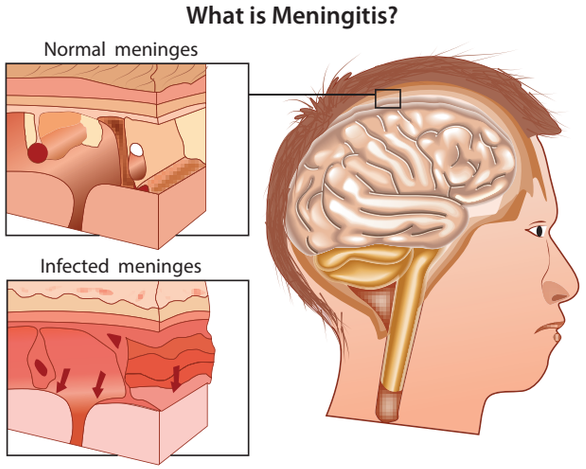


Figure 7.7: Pathogenesis of Meningitis

Table 7.4: Colony morphology of *Neisseria Meningitides* on media

Name of the media	Colony Morphology
Chocolate agar	Colonies are large, colorless to grey opaque colonies.
Mueller Hinton agar	Colonies are small, round, convex grey, translucent with entire edges. The colonies are butyrous in consistency and easily emulsified.

7.5.3 Pathogenesis

N. meningitidis is the causative agent of meningococcal meningitis, also known as pyogenic or septic meningitis. Infection is most common in children

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